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# Data Science

> p-value and hypothesis testing

# What is a hypothesis test?

- Hypothesis tests are used to test the ***validity of a claim*** that is made about a population.

- This claim that's '**on trial**' is called the ***null hypothesis*** (or $\mathbf{H_0}$).

- The ***alternative hypothesis*** ($\mathbf{H_a}$ or $\mathbf{H_1}$) is the one you would believe if the null hypothesis is concluded to be untrue.

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# Hypothesis tests and p-value

- All hypothesis tests use a **p-value** to weight the strength of the evidence (what the data are telling you about the population).
- **$H_0$** or the **null hypothesis** is the thing we are trying to provide evidence against.
- **$H_1$** or the **alternative hypothesis** is what we are trying to prove.
- The null hypothesis, $H_0$ is the **commonly accepted fact**; it is the opposite of the alternative hypothesis.

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# Why is it called Null?

- The word "**null**" in this context means that it's a commonly accepted fact that researchers work to ***nullify*****.** It doesn't mean that the statement is null.

- Perhaps the term should be called the "**nullifiable hypothesis**"

**Eg …**

- Not so long ago, people believed that the world was flat

  **Null hypothesis, $H_0$:** The world is flat.

  **Alternative hypothesis, H1:** The world is round.

- Copernicus, set out to disprove the null hypothesis. This eventually led to the rejection of the null and the acceptance of the alternative hypothesis.

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# How to state the null and alternative hypothesis

Lets look at an example…

- A researcher thinks that if knee surgery patients go to physical therapy twice a week (instead of 3 times), their recovery period will be longer. Average recovery times for knee surgery patients is 8.2 weeks.
- What is accepted is that "**average recovery times for knee patients is 8.2 weeks**". This is the **null** hypothesis.
- The **alternative hypothesis** is that recovery periods take longer than 8.2 weeks – what the researcher is trying to prove. $H_0$ : mean <= 8.2 weeks

$H_1$ : mean > 8.2 weeks

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# Steps to find $H_0$ and $H_1$

- **Step 1**: Figure out your hypothesis from the problem. Lets start with the alternative hypothesis.
- The alternative hypothesis is usually hidden in a problem, and is sometimes a statement of what you expect to happen in an experiment, or research outcome.
- The alternative hypothesis in this example is "**I expect the average recovery period to be greater than 8.2 weeks.**"
- **Step 2**: Convert the alternative hypothesis to maths. Remember that the average can be written as $\mu$.

$$\mathbf{H_1: \mu > 8.2}$$

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# How to state the null hypothesis

- **Step 3**: State what will happen if the alternative hypothesis doesn't come true. This is the **null hypothesis**.

- If the recovery time isn't greater than 8.2 weeks, there are only two possibilities, that the recovery time is equal to 8.2 weeks or less than 8.2 weeks.

$$\mathbf{H_0: \mu \leq 8.2}$$

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# How to state an unclear hypothesis

- What if the alternative hypothesis is unclear?

- Eg A researcher is studying the effects of an exercise program on knee surgery patients. There is a good chance the exercise program will improve recovery time, but there's also the possibility it will make the recovery time worse. Average recovery times for knee surgery patients is 8.2 weeks.
- **Step 1**: State what will happen if the experiment doesn't make any difference.

- That's the null hypothesis - that nothing will happen.

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# How to state an unclear hypothesis

- In this experiment, if nothing happens, then the recovery time will stay at 8.2 weeks.

$$\mathbf{H_0: \mu = 8.2}$$

- **Step 2**: Figure out the alternative hypothesis.

- The alternative hypothesis is the opposite of the null hypothesis. In other words, what happens if our experiment makes a difference?

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# How to state an unclear hypothesis

$$H_1: \mu \neq 8.2$$

# Null hypothesis ($H_0$)

- Null hypothesis states that there is no difference between the groups studied.
- So there is no relationship between the data being studied.
- Example if we were studying a placebo group versus a group with new diabetes medication, then the Null hypothesis states that the blood sugar would be roughly the same for each group.
- Default (null hypothesis) is valid until there is enough evidence to reject this hypothesis.
- In research this may **not** be good – we didn't find the treatment that works, or the risk factor is not as important as first thought.

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# Alternative hypothesis ($H_a$ or $H_1$)

- **Alternative hypotheses** is the opposite of the null hypothesis.
- This hypothesis states there is a difference between groups.
- E.g. there is a relationship between the data variables.
- **Remember** - A researcher wants the alternative hypothesis to be true – discovery of a treatment or a new method that improves patient outcomes

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# What is the p-value?

- The p-value is one of the most important values in statistics and is one of the important outputs in statistical analysis.
- The p-value tells us how likely it is to get a result like this if the NULL hypothesis is **true**.
- Researchers work to **reject** or disprove the null hypothesis.
- Researchers come up with an alternative hypothesis, one that they think explains a phenomenon, and then work to reject the null hypothesis.

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# What is the p-value?

- All hypothesis tests ultimately use a p-value to weigh the strength of the evidence.
- The p-value is a number ***between 0 and 1*** and interpreted in the following way:
- A small **p-value ≤ 0.05** indicates strong evidence against the null hypothesis, so you **reject H0**.
- A large **p-value > 0.05** indicates weak evidence against the null hypothesis, so you **fail to reject H0**.
- p-values very close to the cutoff (0.05) are considered to be marginal (could go either way).

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# Example

- A pizza place claims their delivery times are 30 minutes or less on average but you think it's more than that.
- You conduct a hypothesis test because you believe the null hypothesis is incorrect.

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Therefore …

- $H_0$ : mean delivery time <= 30 minutes
- $H_1$ : mean delivery time > 30 minutes

# Example

- You randomly sample some delivery times and run the data through the hypothesis test, and your p-value turns out to be 0.001, which is much less than 0.05.

- You conclude that the pizza place is wrong; their delivery times are in fact more than 30 minutes on average.

- Of course you could be wrong by having sampled an unusually high number of late pizzas just by chance.

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# Making Conclusions

- To draw conclusions about $H_0$ (reject or fail to reject) based on a p-value, you need to set a predetermined cutoff point where only those p-values less than or equal to the cutoff will result in rejecting $H_0$.

- This cutoff point is called the **alpha level ($\alpha$), or significance level** for the test.

- While 0.05 is a very popular cutoff value for rejecting $H_0$, cutoff points and resulting decisions can vary - some people use stricter cutoffs, such as 0.01, requiring more evidence before rejecting $H_0$, and others may have less strict cutoffs, such as 0.10, requiring less evidence.

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# Making Conclusions

- If $H_0$ is rejected (that is, the p-value is less than or equal to the predetermined significance level), the researcher can say they found a statistically significant result.

- If you get a statistically significant result, you have enough evidence to reject the claim, $H_0$, and conclude that something different or new is in effect (that is, $H_1$).

- The significance level can be thought of as the highest possible p-value that would reject $H_0$ and declare the results statistically significant.

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# Making Conclusions

- Following are the general rules for making a decision about $H_0$ based on a p-value:
- If the p-value is less than or equal to your significance level **($p <= \alpha$)**, then it meets your requirements for having enough evidence against $H_0$; **you reject $H_0$**

- If the p-value is greater than your significance level **($p >= \alpha$)**, your data failed to show evidence beyond a reasonable doubt; **you fail to reject $H_0$**

- However, if you plan to make decisions about $H_0$ by comparing the p-value to your significance level, **you must decide on your significance level ahead of time**.

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# Making Conclusions

- It wouldn't be fair to change your cutoff point after you've got a sneak peak at what's happening in the data.

- You may be wondering if it's okay to say "**Accept $H_0$**" instead of "**Fail to reject $H_0$.**"

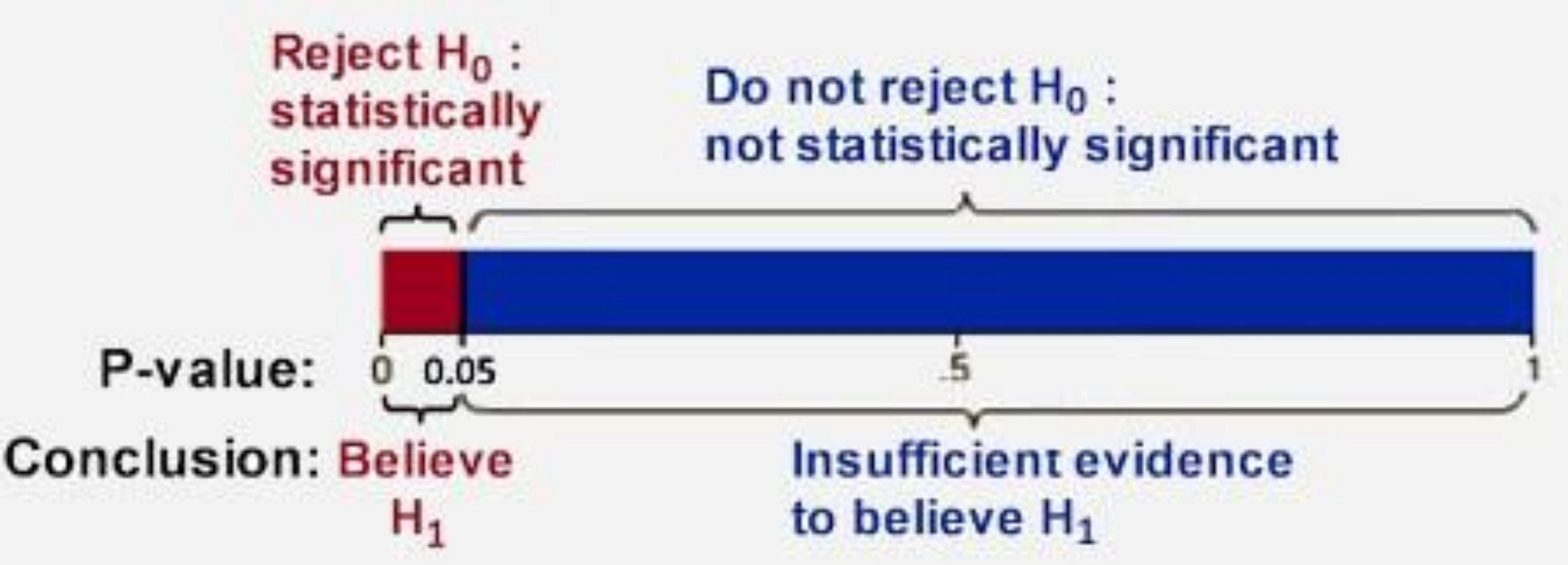
# Setting boundaries for rejecting H0

- These guidelines help you make a decision (**reject or fail to reject $H_0$**) based on a p-value when your significance level is 0.05:
- If the p-value is less than 0.01 (very small), the results are considered highly statistically significant - **reject $H_0$**.
- If the p-value is between 0.05 and 0.01 (but not super-close to 0.05), the results are considered statistically significant - **reject $H_0$**.
- If the p-value is really close to 0.05 (like 0.051 or 0.049), the results should be considered marginally significant - **the decision could go either way**.

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- If the p-value is greater than (but not super-close to) 0.05, the results are considered non-significant - **you fail to reject $H_0$.**

# Summary of boundaries




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- A company claims that its average package delivery time is 2 days.
- A consumer group tests this hypothesis, gets a p-value of 0.04, and concludes that the claim is false: They believe that the average delivery time is actually more than 2 days.
- If the group can stand by its statistics, it has done well to inform the public about the false advertising issue. But what if the group is wrong?

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# Making a false alarm: Type-1 errors

- Even if the group bases their study on a good design, collects good data, and makes the right analysis, it can still be wrong.
- Why? Because its conclusions were based on a sample of packages, not on the entire population.
- Just because the results from a sample are unusual doesn't mean they're impossible.
- A p-value of 0.04 means that the chance of getting your particular test statistic, even if the claim is true.

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# Making a false alarm: Type-1 errors

- You reject $H_0$ in this case because that chance is small. But even a small chance is still a chance.
- Perhaps your sample, though collected randomly, just happens to be one of those samples whose result ended up far from what was expected.
- So, $H_0$ could be true, but your results lead you to a different conclusion.
- How often does that happen? Five percent of the time (or whatever your given cutoff probability is for rejecting $H_0$)

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# Making a false alarm: Type-1 errors

- Rejecting $H_0$ when you shouldn't is called a **type-1 error**.
- In the case of the packages, if the consumer group made a type-1 error when it rejected the company's claim, they created a false alarm.
- To reduce the chance of false alarms, set a low cutoff probability (significance level) for rejecting $H_0$. Setting it to 5% or even better, to 1% will keep the chance of a type-1 error in check.

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# Missing out on a detection: Type-2 errors

- On the other hand, suppose the company really wasn't delivering on its claim.
- Who's to say that the consumer group's sample will detect it?
- If the actual delivery time is 2.1 days instead of 2 days, the difference would be pretty hard to detect.
- If the actual delivery time is 3 days, even a fairly small sample would probably show that something's up.

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# Missing out on a detection: Type-2 errors

- The issue lies with those in-between values, like 2.5 days. If $H_0$ is false, you want to find out about it and reject $H_0$. Not rejecting $H_0$ when you should have is called a **type-2 error** - a missed detection.

- Sample size is the key to being able to detect situations where $H_0$ is false and, thus, avoiding type-2 errors.

# Missing out on a detection: Type-2 errors

- The **more information** you have, the **less variable** your results will be and the more ability you have to zoom in on detecting problems that exist with a claim made by $H_0$.

- This ability to detect when $H_0$ is truly/ false is called the power of a test.
- **Power** is a pretty complicated issue, but what's important for you to know is that the higher the sample size, the more powerful a test.

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# Missing out on a detection: Type-2 errors

- A powerful test has a small chance for a type-2 error

- As a preventative measure to minimize the chances of a type-2 error, statisticians recommend that you select a large sample size to ensure that any differences or departures that really exist won't be missed.

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# Summary of type-1 and type-2 errors

| | Null hypothesis is TRUE | Null hypothesis is FALSE |
|---|---|---|
| **Reject null hypothesis** | Type I Error (False positive) | Correct outcome! (True positive) |
| **Fail to reject null hypothesis** | Correct outcome! (True negative) | Type II Error (False negative) |

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